

## I) Project Goals & Fact-Finding

### 1) Primary Research Question:

- a) What is the total estimated value of the land currently occupied by all golf courses in the United States?

### 2) Core Analytical Framework:

- a) The project will be a cross-sectional analysis, meaning a "snapshot in time" using the most recent data available.
- b) The central concept is a "highest and best use" counterfactual. We are not trying to find the market value of the golf course as a business. Instead, we are estimating what the land would be worth if it were converted to its most likely alternative use.

### 3) Data Strategy (The "Hybrid Model"):

- a) To answer the question, we need a hybrid approach because a single data source is insufficient.
- b) For Urban/Suburban Courses: The "highest and best use" is assumed to be residential development.
- c) For Rural Courses: The "highest and best use" is assumed to be agricultural use.

### 4) Key Data Sources Identified:

- a) FHFA (Federal Housing Finance Agency): This is the primary source for estimating residential land values. It provides data at the county, ZIP, and census tract levels.
- b) USDA (U.S. Department of Agriculture): This is the primary source for estimating agricultural land values. It provides data at the state and regional levels.
- c) BEA (Bureau of Economic Analysis): This source provides the total value of all private land at the state and national levels. Its purpose is not for direct calculation but for providing high-level context to frame the final results.
- d) Golf Course List & Location Data: A master list of all U.S. golf courses with their coordinates (longitude, latitude) is required to link them to the land value data.

### 5) Primary Challenge Identified:

- a) The land value data is priced per acre. Therefore, the most critical missing piece of information is the acreage (land area) for each individual golf course. This must be estimated before the final valuation can be calculated.

## II) Data Sources- Exploration and Selection

### 1) Primary Data Sources (Selected for the Project)

#### a) Source 1: FHFA Residential Land Value Data

- i) Description: An experimental dataset from the Federal Housing Finance Agency estimating the per-acre value of residential land, excluding structure value.
- ii) Methodology: Uses a land residual model on millions of single-family home appraisals submitted to Fannie Mae and Freddie Mac.
- iii) Geographic Granularity: County, ZIP Code, and Census Tract.
- iv) Year/Vintage: The most recent available dataset covers the period from 2012 through 2022.
- v) Role in Project: Primary source for valuing golf course land in urban and suburban areas under a residential "highest and best use" counterfactual.

#### b) Source 2: USDA Agricultural Land Value Data

- i) Description: An annual survey from the USDA's National Agricultural Statistics Service (NASS) summarizing the per-acre market value of agricultural land.
- ii) Methodology: Based on nationwide surveys of farmers and ranchers.
- iii) Geographic Granularity: State and regional.
- iv) Year/Vintage: Published annually; the most recent data available covers 2025.
- v) Role in Project: Secondary source for valuing golf course land in rural areas under an agricultural "highest and best use" counterfactual.

#### c) Source 3: BEA Total Land Value Data

- i) Description: Data from the Bureau of Economic Analysis's Fixed Assets accounts, estimating the total market value of all privately owned land.
- ii) Methodology: Macroeconomic estimation based on a variety of national and state-level economic data.
- iii) Geographic Granularity: State and National.
- iv) Year/Vintage: Updated regularly; data is available through the most recent completed year.
- v) Role in Project: Contextual data only. Used to frame the final results (e.g., what percentage of a state's total land value is represented by golf courses). Not used for direct calculation.

#### d) Source 4: U.S. Golf Course Location Data

- i) Description: A comprehensive list of U.S. golf courses.
- ii) Methodology: Compiled from various public online directories (e.g., GolfLink, which licenses data from the National Golf Foundation).
- iii) Geographic Granularity: Individual course level, with latitude and longitude coordinates.
- iv) Year/Vintage: Assumed to be current as of the date of compilation.
- v) Role in Project: The master list of subjects for the analysis, providing the link between a specific course and its geographic location.

## 2) Data Sources Considered but Not Used (and Why)

### a) Source 1: County Tax Assessor / Parcel Data

- i) Description: The official assessed value of land for individual property parcels, available from individual county government websites.
- ii) Reason Not Used for Primary Analysis: While this is the most granular and "official" source for a specific parcel, there is no centralized national database. Compiling this data for all ~15,000+ golf courses would require manually visiting and scraping over 3,000 individual county websites, which is not feasible for a national-scale project.
- iii) Potential Future Use: Excellent for validation and case studies. Could be used to spot-check the accuracy of the FHFA/USDA model for a small sample of courses.

### b) Source 2: Commercial Real Estate Data Providers (e.g., CoreLogic, Zillow ZTRAX)

- i) Description: Private companies that compile and sell comprehensive, parcel-level real estate data, including land value estimates.
- ii) Reason Not Used: Access to these datasets is prohibitively expensive, typically requiring institutional licenses costing thousands of dollars. This project is limited to publicly available data.
- iii) Potential Future Use: Would be the "gold standard" data source if funding were available.

### iv) Source 3: Fannie Mae / Freddie Mac Raw Data

- v) Description: The raw appraisal and mortgage data that the FHFA uses to create its land value estimates.
- vi) Reason Not Used: This data is not publicly available. It is proprietary and contains sensitive personal and financial information. The FHFA dataset is the only public, aggregated version of this data.

### III) Data Cleaning, Harmonization, and Enrichment

#### 1) Initial Data Cleaning and Standardization

a) Objective: To process the raw, disparate source files into a consistent and usable format.

b) Key Actions:

- i) Master Golf Course List: Describe the process of parsing the messy, single-string data into discrete columns (course\_name, street\_address, city, state, zip\_code, course\_type, holes). Mention the use of regular expressions and the manual correction loop for rows that failed to parse automatically.
- ii) Economic & Land Data: Detail the steps taken to harmonize the supporting files.
  - Handling Complex Headers: Note the strategy of renaming columns by position to handle multi-line or merged headers in the Excel files (HPI, Land Value).
  - Standardizing Geography: Mention converting full state names to 2-letter abbreviations to ensure they could be joined correctly across all datasets.
  - Cleaning Data Types: Describe the process of converting text-based numbers (e.g., currency with "\$" and ",") into a numeric format and cleaning the holes column from "18 Holes" to just 18.

#### 2) Primary Data Enrichment: Acreage Estimation

a) Objective: To create the most critical variable for the analysis—the land area (acreage) for each golf course—which was not present in the original data.

b) Methodology: A hybrid, "best-available-data" approach was used, combining two independent, publicly available spatial data sources.

i) Source 1: U.S. Census Bureau (tigris package)

- Process: Downloaded official "Area Landmark" polygons for golf courses. Used a spatial "nearest neighbor" join with a distance threshold to match course coordinates to these official polygons.
- Outcome: Provided high-reliability acreage data but had a low match rate (~9%) due to the incompleteness of the Census landmark data.

ii) Source 2: OpenStreetMap (osmdata package)

- Process: Programmatically queried the OpenStreetMap API for leisure=golf\_course polygons near each course's coordinates. Mentioned managing API rate limits with a Sys.sleep() command.
- Outcome: Provided medium-reliability, crowd-sourced data but achieved a much higher match rate.

iii) Final Combination:

- Process: The two acreage estimates were combined using a coalesce function. This prioritized the high-reliability tigris data and only used the osmdata value if the tigris value was missing.
- Data Provenance: A new column, acreage\_source, was created to track the origin of every acreage value ("Tigris," "OSM," or "Imputed"), ensuring transparency.
- Final Match Rate: This hybrid approach successfully estimated the acreage for ~62% of the courses.

### 3) Handling Missing Data: Multiple Imputation (MICE)

a) Objective: To create robust, statistically valid estimates for the ~38% of courses still missing an acreage value after the enrichment phase.

#### b) Methodology:

i) Technique Chosen: Multiple Imputation by Chained Equations (MICE), generating 5 complete datasets ( $m=5$ ).

ii) Justification: Explain why MICE was chosen over simpler methods (like mean/median imputation). It is the gold standard for academic research because it accounts for the uncertainty of the imputation process, leading to valid statistical inferences.

iii) Model Predictors: List the rich set of variables used to make the imputed estimates as accurate as possible: holes, course\_type, longitude, latitude, state, and state\_land\_value.

iv) Validation: State that the success of the imputation was confirmed with diagnostic plots, which showed excellent model convergence and distributional validity.

#### • Collinearity:

⇒ The MICE algorithm flagged collinearity between the spatial predictors (longitude/latitude) and the regional predictor (state). This resulted in "logged events" where redundant variables were temporarily dropped during iteration. This was expected behavior for geographic data and did not negatively impact the model's convergence.

## IV) Analytical Framework and Methodology

### 1) Implementation of the Hybrid Valuation Model

- a) Objective: To calculate the estimated "highest and best use" land value for every golf course in the master dataset.
- b) Step 1: Classifying County Location Type:
  - i) Method: Use the official USDA Rural-Urban Continuum Codes (RUCC) to classify every U.S. county as either "Urban" or "Rural."
  - ii) Process: Join the RUCC dataset to the master golf course file using 5-digit County FIPS codes to ensure accurate matching (avoiding errors with duplicate county names like "Washington" or "Lincoln").
  - iii) Process: Join the RUCC dataset to the master golf course file. A defensible rule will be applied (e.g., RUCC codes 1-3 are classified as "Urban," and codes 4-9 as "Rural"). This creates a new county\_type variable for each course.
- c) Step 2: Conditional Value Calculation:
  - i) Method: A conditional calculation will be performed based on the county\_type.
  - ii) Process:
  - iii) If county\_type is "Urban", the value will be calculated as:  $\text{final\_acreage} * \text{fhfa\_price\_per\_acre}$ .
  - iv) If county\_type is "Rural", the value will be calculated as:  $\text{final\_acreage} * \text{usda\_price\_per\_acre}$ .
- d) Outcome: This will produce the primary analytical variable: estimated\_land\_value for every course.

### 2) Descriptive and Summary Analysis

- a) Objective: To generate the headline findings and key summary statistics from the calculated data.
- b) Key Metrics to be Generated:
  - National Aggregate Value: The sum of estimated\_land\_value for all courses to answer the primary research question.
  - Breakdown by State: Summary tables showing the total value and number of courses for each state.
  - Breakdown by Course Type: A critical comparison of the total land value held by Private vs. Public vs. Municipal courses.
  - Breakdown by Metropolitan Statistical Area (MSA): Aggregating the data to the MSA level to identify which urban centers have the most capital tied up in golf course land.

### 3) Econometric and Spatial Analysis

a) Objective: To explore the drivers of land value and visualize the spatial distribution of the findings.

i) Method 1: Regression Analysis:

- Purpose: To understand the factors that are correlated with a golf course's underlying land value.
- Model Specification: A regression model will be estimated with `estimated_land_value` as the dependent variable and predictors such as `holes`, `course_type`, and regional indicators.
- MICE Protocol: The analysis will be conducted using the proper MICE protocol: the regression will be run on each of the 5 imputed datasets, and the results will be combined using the `pool()` function to generate statistically valid coefficients and standard errors.

b) Method 2: Spatial "Hotspot" Mapping:

- Purpose: To visually identify geographic concentrations of high-value golf course land.
- Process: The `estimated_land_value` data will be aggregated to the county level. A choropleth map of the U.S. will be created, with counties shaded according to the total value of golf course land they contain.

### 4) Contextualization of Final Results

a) Objective: To frame the significance of the aggregate findings using the BEA data.

b) Process: The final aggregate land value (e.g., for a specific state or for the entire nation) will be compared to the total value of all private land as reported by the Bureau of Economic Analysis (BEA).

c) Example Outcome Statement: "The aggregate estimated value of golf course land in California is \$X billion, representing approximately Y% of the total \$Z trillion in private land value for the entire state as estimated by the BEA."

## V) Discussion of Alternative Land Uses and Limitations

### 1) Zoning and Permitted Uses (The "Real World" Complexity)

- a) Objective: To acknowledge that while the study uses a binary "Residential vs. Agricultural" model, the actual legal and physical potential of the land is often more complex.
- b) Insight: Professional appraisal data indicates that golf course land often falls under specific zoning designations (e.g., P-2 Preservation in Hawaii) that allow for a wide range of alternative uses beyond simple farming or housing.
- c) Examples of Permitted Uses:
  - i) Agricultural/Rural: Livestock, forestry, crop production, game preserves, and aquaculture.
  - ii) Industrial/Utility: Bio-fuel processing facilities, waste disposal transfer stations, and utility installations (Type A/B).
  - iii) Institutional: Cemeteries, hospitals, prisons, and universities.
  - iv) Recreational: Vacation cabins, outdoor education facilities, and marinas.
- d) Validation of the USDA Proxy: Despite this variety, many of the permitted rural uses (forestry, livestock, crop production) align closely with the USDA Agricultural Land Value data. This validates the decision to use agricultural value as a broad proxy for the "highest and best use" of rural golf course land, even if the specific use varies by parcel.

### 2) Tax Incentives and Policy Distortions

- a) Objective: To discuss how tax policy can create a gap between the potential market value of the land and its current use.
- b) Key Example (The "New Jersey Loophole"):
  - i) Scenario: In states like New Jersey, land can be designated as a "cemetery" even if only a small portion is used for interments (e.g., a single grave).
  - ii) Impact: This designation can exempt the property from real estate taxes, creating a massive financial incentive to maintain the land as a golf course rather than developing it for residential use.
- c) Implication for the Study: This study estimates the economic opportunity cost of the land (what it could be worth). However, tax loopholes act as a barrier to realizing this value, effectively subsidizing the golf course's existence.

### 3) Environmental Externalities

- a) Objective: To acknowledge that the economic valuation does not account for environmental costs.
- b) Key Issue: Golf courses are often sources of significant pollution, including fertilizer runoff and physical debris (e.g., decaying golf balls in coastal areas like Monterey/Santa Cruz).
- c) Limitation: This study focuses purely on market value. It does not subtract the "negative external costs" of pollution, nor does it add the "positive external benefits" of green space or carbon sequestration. A full "social cost-benefit analysis" would require quantifying these environmental factors, which is outside the scope of this project.